

Jasper Sands

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Quantum error correction, quantum algorithms, and machine learning for quantum-device characterization. MS in Computer Science at Columbia. Thesis at Diraq and UNSW Sydney on automating silicon spin-qubit measurement.

EDUCATION

Columbia University, MS in Computer Science, Quantum Computing Track · GPA 3.95/4.0 Sep 2025 – May 2027

Quantum coursework: Error Correction · Engineering · Simulation & Computing Lab · Optimization & ML.

TA: Introduction to Quantum Computing.

Washington University in St. Louis, BS in Computer Science and Economics · GPA 3.81/4.0

Aug 2021 – May 2025

Coursework: Machine Learning · Analysis of Algorithms · Computer Security. **TA:** Malware Analysis; Parallel Programming.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for financial Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026

- Wrote the GPU-accelerated state prep, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation.
- Measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ quasi-Monte Carlo classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XZZX surface codes, with exact MWPM and Union-Find decoding client-side.
- Finite-size scaling estimates thresholds of 3.2% with MWPM, 2.7% with Union-Find, 14.7% at code capacity.

Q-Search: proof-gated quantum algorithm research and verification engine qsearch.jaspersands.com

- Automated search for structural quantum advantage on non-abelian HSP, dihedral coset, and linear code equivalence problems.
- Formal proof gates test each mechanism against 1,200+ classical baselines across 720+ theorem modules, indexing negative results.

LLM-Based Lossless Text Compression · *PyTorch, CUDA* jaspersands.com/llmcompression

- Arithmetic coding on LLM predictions, 5–16× average and 39× best case, at 100–1000× lower throughput than gzip and zstd.

RESEARCH

Diraq / UNSW Sydney · *MS Thesis Research*

Sep 2026 – May 2027

- Automating silicon spin-qubit characterization under Dr. Nard Dumoulin Stuyck, replacing manual tuning and interpretation.
- Acquiring charge measurements from cryogenic probing, and classifying device states against what an expert would call by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher*

Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and optimizing precision under a fixed false-positive budget.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher*

May – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian, grounded in molecular graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher*

Sep 2024 – May 2025

- Fine-tuned LLaMA-3 with Hugging Face for QA over 50k foster-care policy documents, with human-in-the-loop feedback.

EXPERIENCE

Smack Technologies · *AI Intern*

May – Aug 2026

- Built and expanded physics and sensor simulation, integrating the knowledge graph to widen corpus generation.
- Benchmarked open-weight models head to head on speed, intelligence and concurrency, speeding corpus generation 64x.
- Bounded simulator error to 0.55% against references outside its own tests, in a report that regenerates when the physics changes.

Highnote · *Cybersecurity Engineer Intern*

May – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day, and wrote 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern*

May – Jul 2023

- Built internal admin and secure-query tooling in JavaScript and Rails via Retool, saving roughly 250 engineering hours a month.

COMPETITIONS & LEADERSHIP

- **Overall Winner & NVIDIA Ecosystem Award**, MIT iQuHACK 2026, qualified for the NYU Abu Dhabi Quantum and AI Hackathon 2027 · **Audience Favorite**, Harmoniqs Quantum Design 2026 · **Runner-Up**, Qualcomm Snapdragon Hackathon 2026
- **Graduate Lead**, Columbia Quantum Algorithms Reading Group: weekly sessions through all 33 chapters of Andrew Childs' notes.

SKILLS

Quantum tools: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP

Quantum methods: Hamiltonian simulation, phase and amplitude estimation, QSVT and block encoding, VQE, QAOA · surface codes, stabilizer formalism, MWPM and Union-Find decoding, threshold estimation, noise modeling

Languages: Python, C++, Rust, C, MATLAB, JavaScript, TypeScript, Go, Java

ML & systems: PyTorch, CUDA, Hugging Face · Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, LaTeX